

EEE412 Microwave Electronics- LAB 1 Experimental Work

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"I completed this homework assignment independently."

Part 1

Experimental Setup

Two 5 m RG-58 coaxial cables were connected in series, with a 100 resistor at the generator side. The Rigol DG5102 function generator produced a 5 V, 10 kHz signal, and a DSO5052A oscilloscope was connected at the midpoint through a BNC T-junction. The far end of the cable was successively left open, connected to a 50 ohm load, and short-circuited. Measurements were first made in Square Wave mode (step-like input) and then repeated in Pulse mode (4 ns width).

a) The end is left open with a square wave with %50 duty cycle.

An open has reflection coefficient $\Gamma_L = +1$, so the first echo adds to the incident step, raising the voltage. If the source/scope input isn't perfectly matched ($\Gamma_S \neq 0$, e.g., Hi-Z input or not exactly 50 Ω), the wave bounces back and forth between the source and the open; each round-trip a fraction reflects again and adds, producing the discrete upward steps spaced by the line's round-trip delay.

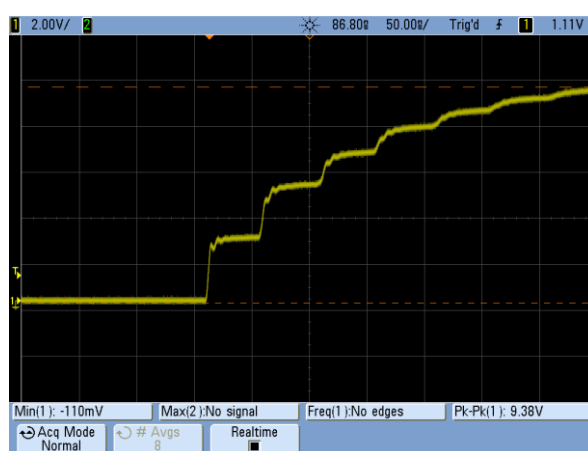


Figure 1: Square Wave with %50 Duty Cycle, Open end Result

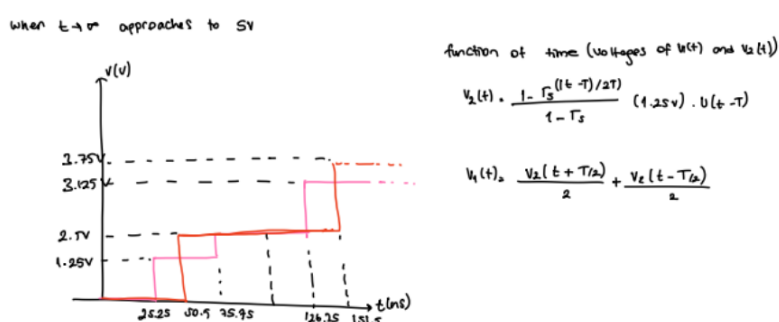


Figure 2: Lab1 Pre, Square wave Coming to the Open end of the Transmission Line

b) Square wave with %50 duty cycle 50Ω load at the open end

Matching is observed with 50Ω load, with a very low reflection.

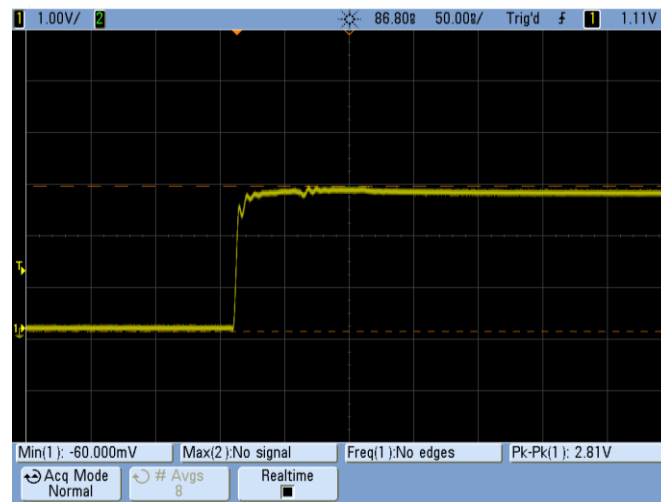


Figure 3: Square Wave with %50 Duty Cycle, 50Ω Load Output Matching Result

c) Open end is short circuited with again a square wave with %50 duty cycle.

With a short circuit end reflection occurs with a inverted voltage value. When you short-circuit the open end of a transmission line, the reflected wave is inverted because a short has a reflection coefficient of -1 ($\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$). This causes the voltage at the measurement point to dip below its initial level when the reflection returns after one round-trip delay.

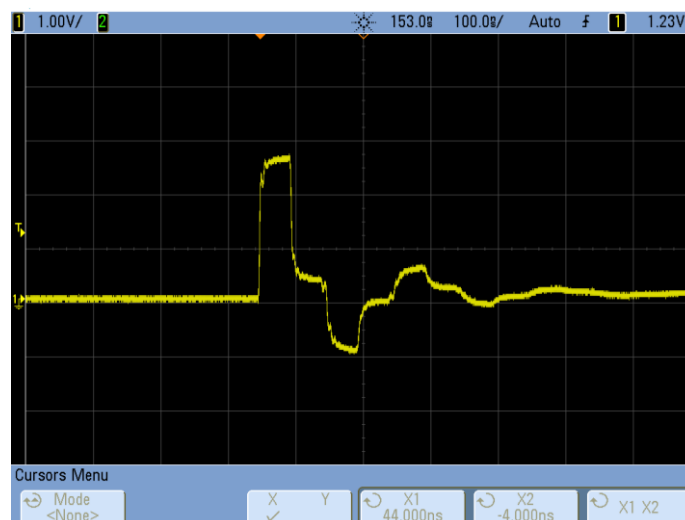


Figure 4: Square Wave with %50 Duty Cycle, Short Circuited Output Result

a) The end is left open with a pulse coming with 4ns pulse width.

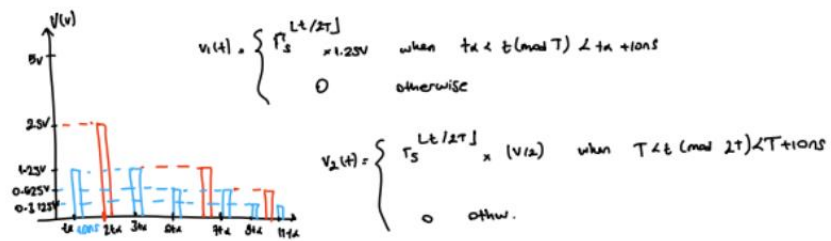


Figure 5: Lab1 Pre Calculations and Pulse Coming to Open Circuit end Transmission Line

See below Figure is matching with the Pre-lab results

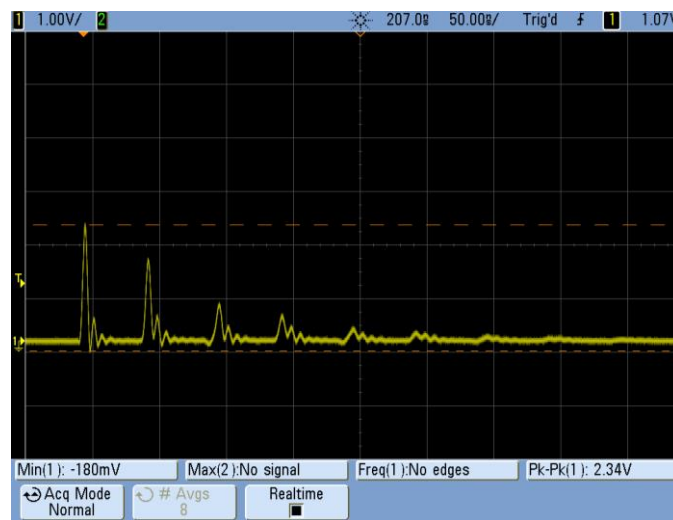


Figure 6: Open End with a Pulse Coming with 4ns Width

b) Pulse coming with 4ns pulse width to the 50Ω load at the output

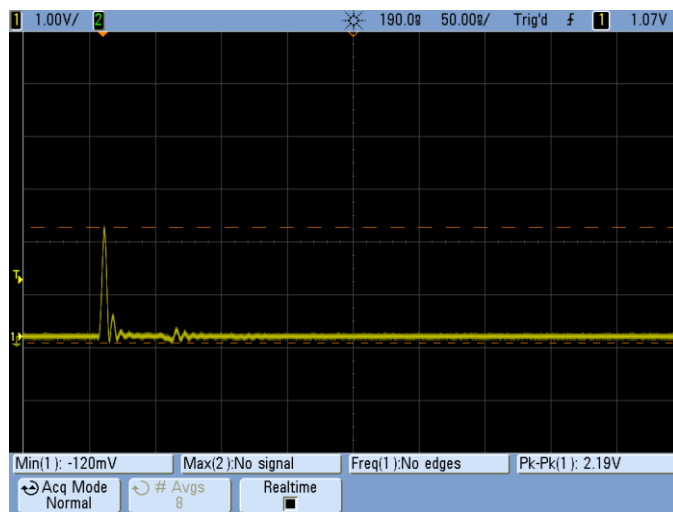


Figure 7: 4ns Pulse Width to the 50Ω Load at the Output

c) Open end is short circuited with again a pulse



Figure 8: 4ns Pulse Width to the Short Circuited Output

Part 2

2.1) Without tuning with the Electrical length values calculated in the preliminary work of this lab we reach about 263MHz (Figure 10) which is the minimum of the S(1,1) value we observed via ADS, and also in the lab. See also, Figure without the proper tuning operation (length manipulation) 280MHz is a little away from the center of the Smith Chart. See below Figure for the initial calculated lengths of the transmission lines.

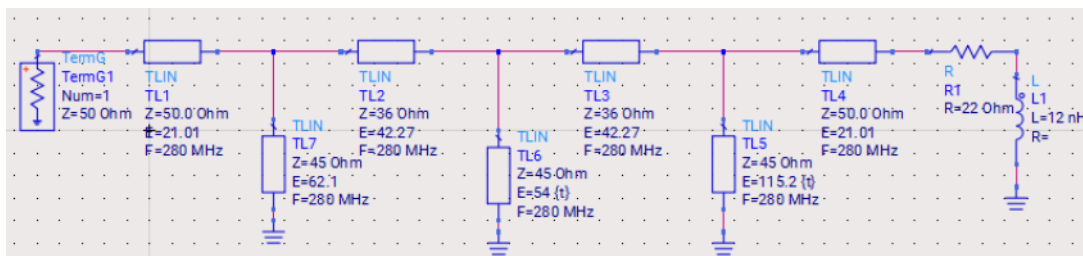


Figure 9: Pre Lab Transmission Line Matching Circuit

$$\lambda_{line} = \frac{c}{f} = \frac{3 \times 10^8}{280 \times 10^6} = \frac{300}{280} \approx 1.07 \text{ m}$$

$$E = 360^\circ \frac{l}{\lambda_{line}}$$

Note that we are working with free air (no E calculation needed)

since 360° corresponds to 1.07 m
below electrical lengths corresponds to

$$62.1^\circ \rightarrow 12.9 \text{ cm (for TL7)}$$

$$54^\circ \rightarrow 11.85 \text{ cm (for TL6)}$$

$$115.2^\circ \rightarrow 23.968 \text{ cm (for TL5) (approximately)}$$

with these values transmission line lengths are fixed!

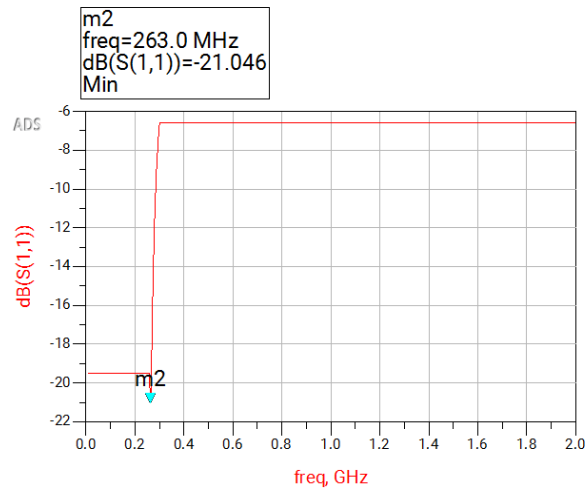


Figure 10: $S(1,1)$ Result Before Tuning

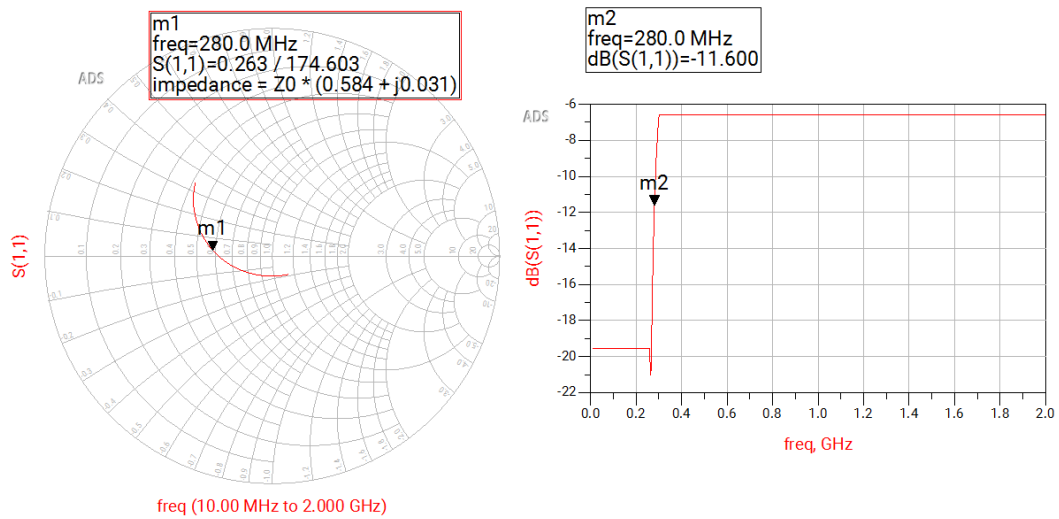


Figure 11: $S(1,1)$ Results Before Tuning

2.2)

With the arm (triple stub tuner) length adjustments we derive the following Smith Chart and observed that our $S(1,1)$ parameter is centered at the frequency 280MHz (my center frequency), meaning that nearly zero reflection occurs at that frequency (see below Figure).

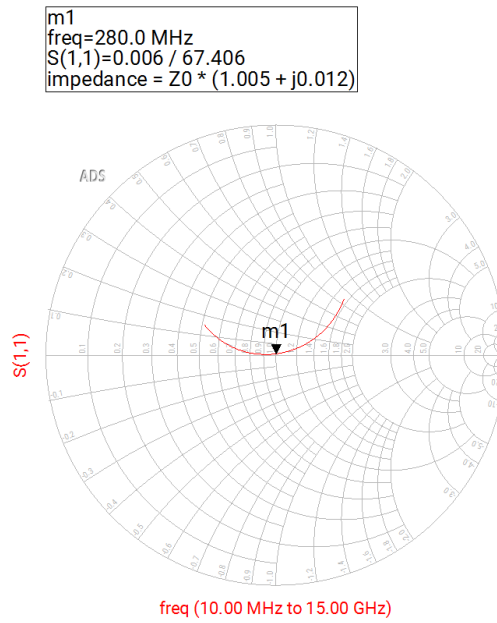


Figure 12: $S(1,1)$ with 280MHz is Centered (Perfect Matching) After Tuning

New approximate lengths after tuning are below:

- 12.9 cm ➡ 18.45 cm (TL 7)
- 7.185 cm ➡ 16 cm (TL 6)
- 23.968 cm ➡ 34.28 cm (TL 5)



Figure 13: New Approximate Lengths After Tuning

Part 3

In this part we observed a LowPass Filter see below Figures for the impedance matching for low frequency values. See $S(1,1)$ sees the min value for low frequencies in Figure 14 And see $S(2,2)$ for the filter characteristics (Figure 14).

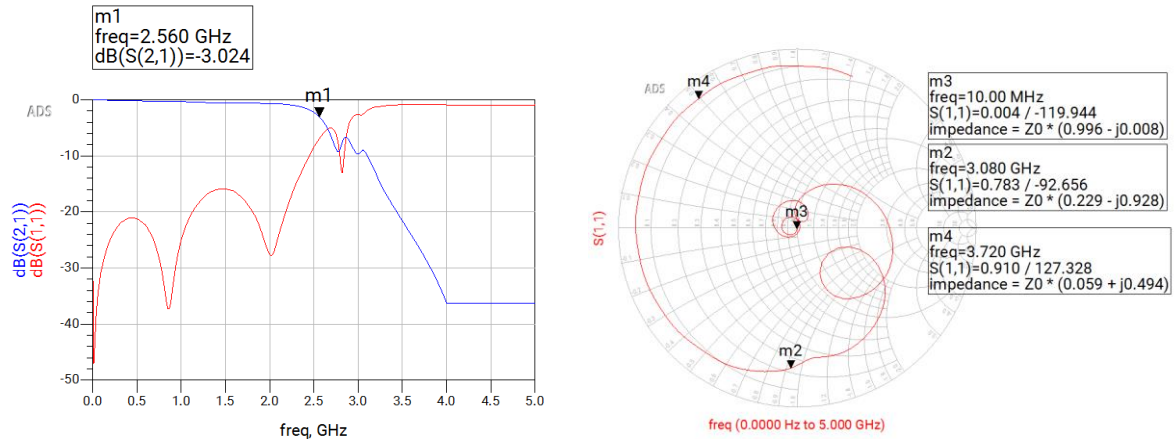


Figure 14: $S(1,1)$ and $S(2,2)$ Result of the LowPass Filter

Part 4

In this part we observed a BandPass Filter see below Figures for the impedance matching for the center frequency 870MHz. See $S(1,1)$ sees the min value for the center frequency in Figure 15 And see $S(2,2)$ for the filter characteristics (Figure 15).

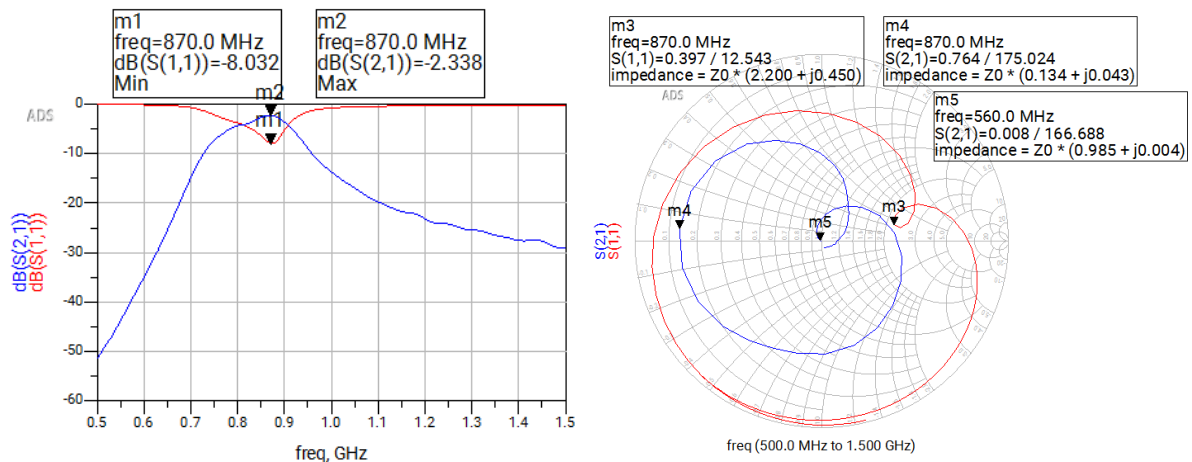


Figure 15: $S(1,1)$ and $S(2,2)$ Result of the BandPass Filter